

## CDE Loss — SDSS

Parametric

Nonparametric

Foundation

| n    | Parametric       |                    |                  |                  |                  |                  |                     |                       |                  |                      |                        |                  | Nonparametric    |                  |                  |                  |                   |                  |                  |               | Foundation        |                   |                   |
|------|------------------|--------------------|------------------|------------------|------------------|------------------|---------------------|-----------------------|------------------|----------------------|------------------------|------------------|------------------|------------------|------------------|------------------|-------------------|------------------|------------------|---------------|-------------------|-------------------|-------------------|
|      | LinearGauss-Homo | LinearGauss-Hetero | Student-t        | LogNormal-Homo   | LogNormal-Hetero | Gamma-GLM        | LinGauss-Homo-Ridge | LinGauss-Hetero-Ridge | Student-t-Ridge  | LogNormal-Homo-Ridge | LogNormal-Hetero-Ridge | Gamma-GLM-Ridge  | FlexCode-RF      | FlexZBoost       | BART-Homo        | BART-Hetero      | Flow-Spline       | Quantile-Tree    | CatMLP           | MDN           | TabPFN 2.5        | RealTabPFN 2.5    | TabICL Quantiles  |
| 500  | -4.7<br>(0.1)    | -5.4<br>(0.3)      | -4.8<br>(0.2)    | -4.3<br>(0.2)    | -4.2<br>(0.1)    | -4.2<br>(0.1)    | -4.7<br>(0.1)       | -5.4<br>(0.2)         | -4.8<br>(0.2)    | -4.3<br>(0.2)        | -4.3<br>(0.1)          | -3.1<br>(0.1)    | -5.3<br>(2.7e-2) | -5.2<br>(0.1)    | -3.5<br>(5.6e-2) | -3.2<br>(0.2)    | -5.2<br>(0.3)     | -2.0<br>(0.5)    | -4.4<br>(0.2)    | -6.1<br>(0.4) | -8.9<br>(0.3)     | -8.9<br>(0.3)     | -8.9<br>(0.3)     |
| 1K   | -4.8<br>(0.1)    | -5.9<br>(0.1)      | -5.1<br>(0.2)    | -4.4<br>(0.1)    | -4.5<br>(9.0e-2) | -4.3<br>(0.1)    | -4.8<br>(0.1)       | -5.9<br>(0.1)         | -5.1<br>(0.2)    | -4.4<br>(0.1)        | -4.5<br>(8.2e-2)       | -3.2<br>(5.9e-2) | -5.9<br>(0.1)    | -6.0<br>(0.2)    | -4.3<br>(4.0e-2) | -4.0<br>(0.2)    | -7.5<br>(0.3)     | -4.1<br>(0.2)    | -6.1<br>(0.4)    | -7.4<br>(0.3) | -10.2<br>(0.3)    | -10.2<br>(0.3)    | -10.2<br>(0.3)    |
| 10K  | -4.5<br>(1.5e-2) | -5.5<br>(2.8e-2)   | -4.8<br>(2.2e-2) | -3.9<br>(3.7e-2) | -4.1<br>(2.2e-2) | -3.9<br>(3.7e-2) | -4.5<br>(1.5e-2)    | -5.5<br>(2.8e-2)      | -4.8<br>(2.2e-2) | -3.9<br>(3.7e-2)     | -4.1<br>(2.2e-2)       | -3.1<br>(2.3e-2) | -6.7<br>(3.6e-2) | -6.9<br>(5.5e-2) | -6.0<br>(4.5e-2) | -4.6<br>(0.1)    | -9.4<br>(0.2)     | -6.4<br>(6.0e-2) | -8.5<br>(0.1)    | -8.6<br>(0.3) | -10.8<br>(9.6e-2) | -10.8<br>(9.7e-2) | -10.8<br>(9.8e-2) |
| 50K  | -4.6<br>(9.7e-3) | -5.5<br>(1.8e-2)   | -4.9<br>(1.8e-2) | -4.0<br>(2.9e-2) | -4.2<br>(1.3e-2) | -4.0<br>(2.9e-2) | -4.6<br>(9.6e-3)    | -5.5<br>(1.8e-2)      | -4.9<br>(1.8e-2) | -4.0<br>(2.9e-2)     | -4.2<br>(1.3e-2)       | -3.2<br>(1.9e-2) | -7.0<br>(1.6e-2) | -7.4<br>(2.0e-2) | -6.7<br>(1.4e-2) | -5.6<br>(4.5e-2) | -10.0<br>(9.8e-2) | -7.7<br>(0.2)    | -9.6<br>(6.5e-2) | -9.1<br>(0.2) | -10.9<br>(4.3e-2) | -10.8<br>(3.8e-2) | —                 |
| 100K | -4.5<br>(7.7e-3) | -5.5<br>(1.3e-2)   | -4.9<br>(1.7e-2) | -4.0<br>(1.1e-2) | -4.1<br>(7.2e-3) | -4.1<br>(1.3e-2) | -4.5<br>(7.7e-3)    | -5.5<br>(1.3e-2)      | -4.9<br>(1.7e-2) | -4.0<br>(1.1e-2)     | -4.1<br>(7.2e-3)       | -3.2<br>(9.8e-3) | -7.0<br>(9.4e-3) | -7.5<br>(1.6e-2) | -6.8<br>(1.6e-2) | -5.8<br>(4.0e-2) | -10.2<br>(0.1)    | -8.0<br>(0.2)    | -9.7<br>(0.2)    | -8.8<br>(0.2) | —                 | —                 | —                 |
| 250K | -4.6<br>(2.5e-3) | -5.5<br>(4.1e-3)   | -4.9<br>(6.2e-3) | -4.1<br>(4.7e-3) | -4.2<br>(6.7e-3) | -4.1<br>(7.2e-3) | -4.6<br>(2.5e-3)    | -5.5<br>(4.1e-3)      | -4.9<br>(6.2e-3) | -4.1<br>(4.7e-3)     | -4.2<br>(6.7e-3)       | -3.2<br>(6.9e-3) | -7.0<br>(6.8e-3) | -7.6<br>(1.3e-2) | -7.0<br>(9.3e-3) | —                | -10.3<br>(6.2e-2) | —                | -9.9<br>(0.3)    | -9.4<br>(0.1) | —                 | —                 | —                 |
| 500K | -4.6<br>(2.9e-3) | -5.5<br>(6.3e-3)   | -4.9<br>(6.3e-3) | -4.1<br>(4.4e-3) | -4.2<br>(2.1e-3) | -4.1<br>(4.5e-3) | -4.6<br>(2.9e-3)    | -5.5<br>(6.3e-3)      | -4.9<br>(6.3e-3) | -4.1<br>(4.4e-3)     | -4.2<br>(2.1e-3)       | -3.2<br>(2.9e-3) | -7.0<br>(6.3e-3) | -7.6<br>(1.4e-2) | -7.1<br>(9.2e-3) | —                | -10.5<br>(4.2e-2) | —                | -10.1<br>(0.1)   | -9.2<br>(0.1) | —                 | —                 | —                 |